

Thermostat Setbacks—Do They Really Work?

During the heating and cooling seasons of 2002–2003, the Canadian Centre for Housing Technology (CCHT) ran a series of trials to determine actual energy savings from thermostat setback and setforward.

by **Marianne Armstrong**

Temporarily adjusting the temperature setting on the thermostat at night, or while occupants are away from home, seems to be a simple and easy way to save energy in the heating and cooling seasons. Many homeowners set back the thermostat (reducing the set temperature) during the night, as well as during the workday, by means of a conventional thermostat or with the aid of a programmable model. In summer, a similar strategy can be employed by setting forward (increasing the set temperature) during the workday, reducing the load on the A/C system during peak hours. But do setback and setforward strategies really help save energy?

The Twin Research House facility at the Canadian Centre for Housing Technology (CCHT) was built in 1998. It consists of two identical two-story houses built to R-2000 standards by a local Ottawa builder. Features of these houses include low-e argon-filled windows and a simulated occupancy system. (For more on the Twin House facility, see “Do CFLs Save Whole-House Energy?” p. 20). During the winter heating season of 2002–2003, the CCHT, where I work, ran a series of trials in the twin houses—one referred to as the reference house and the other the test house—to determine actual energy savings from thermostat setback, and to examine the resulting house temperatures and recovery times.

Three different winter setback settings were examined and compared to the benchmark, which was 72°F (22°C). These settings were 64°F (18°C) night setback; 64°F (18°C) day and night setback; and 61°F (16°C) day and night setback. We also performed a set of summer trials to determine the effect of thermostat setting on A/C performance, and we examined summer indoor humidity levels.

Data Collection

Two modified gas meters with a pulse output connected to the main data acquisition system (DAS) monitored the gas consumption of the furnace. The total gas consumption data were collected at five-minute intervals. Two electric pulse meters measured furnace and A/C electrical consumption. These data were collected at five-minute intervals by the DAS. Furnace fan on time was measured, indicating the total amount of time the furnace fan circulation motor ran at high speed (heating or cooling mode), as opposed to circulation speed. On-time data were collected by another DAS in place to monitor transient performance of mechanical equipment at much shorter time intervals—10 seconds.

In addition to measuring energy consumption, we wanted to understand

the overall effect of thermostat setback on the house and on the occupants’ needs. During both summer and winter trials, we examined house floor temperatures (basement, ground floor, and second floor). We examined window and drywall surface temperatures during winter experiments only, in order to determine condensation risks.

We chose five days for temperature analysis in each of the thermostat trials in the twin houses. Five consecutive days were not available in all cases. Instead, days were chosen to include the coldest possible outdoor temperature in winter or the hottest possible outdoor temperature in summer, when outdoor temperature would have the greatest effect on energy use inside.

Changes in basement, ground floor, and second-floor air temperatures were tracked using thermocouples in key locations in both houses to help us to understand the overall effect of



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thermostat setback. These temperatures were measured at a point half way up the wall. The thermostat itself is located in the hallway of the ground floor, beside the ground floor thermocouple.

Drywall surface temperatures were also measured to ensure that these temperatures did not approach the dew point of the surrounding air, which would lead to condensation problems (see Figure 1). Air at 72°F and 30% humidity will condense on a surface with a temperature below 38.7°F (3.7°C).

Window temperature measurements were also important in our analysis. If the temperature of the glass or the window frame drops below the dew point of the ambient air, condensation (or ice) will form and may cause water damage to the surrounding wall. All windows in the CCHT houses are argon-filled double-pane units with a low-e coating. Window surface temperatures were measured at five different locations on each window (see Figure 2). Three different windows were examined: in bedroom 2, a second floor south facing window; in the living room, a ground floor north facing



The Twin Research House facility consists of two identical two-story houses.

window; and in the dining room, a ground floor south facing window.

The ground floor thermocouple, located beside the thermostat, was used as the basis for recovery time calculations. The temperature of this thermocouple is captured at five-minute intervals by the main DAS. This gives a better resolution for calculating recovery time than other thermocouples in the house, whose temperatures are recorded hourly. The benchmark condition was examined to determine the relationship between the daily average ground floor temperature of the reference house and that of the test house. This ground floor temperature correlation was then used to determine the expected average ground floor temperature of the test house for any given day during the setback and setforward trials. This expected average was set as the threshold temperature for determining recovery time. Recovery time was calculated from the moment when the

thermostat automatically reset to 72°F to the moment when the house reached the threshold temperature.

Humidity Measurements

Humidity is an important factor in determining occupant comfort. During the winter setback experiments, relative humidity (RH) in the houses was very low (about 10%). This was because there were no real occupants, and no humidifiers were run. In the summer, water is removed from the air in the form of condensation on the indoor air conditioner coil. During the summer trials, this condensation was collected and measured by means of a tipping scale.

The RH of each floor is recorded hourly by the main DAS. RH measurements in conjunction with hourly temperature measurements were used to calculate the humidity ratio of air (grams of vapor per kilogram of air) for each floor of the house. The three humidity ratios—for basement, ground floor, and second floor—were then

averaged to generate an average house humidity ratio. This allowed us to compare the moisture content of the test house with that of the reference house.

Outdoor temperature and humidity were measured and recorded every five minutes by means of a thermocouple and humidity sensor mounted on the exterior of the reference house. During the summer experiments, solar radiation incident on the south wall of the reference house was measured at five-minute intervals by a wall-mounted pyronometer. These data were then used to separate cloudy days from sunny days during the setforward experiment. A total vertical solar radiation of 8.5 megajoules per square meter (MJ/m²) per day was arbitrarily chosen to divide cloudy days from sunny days.

At the time of the winter setback experiment, the pyronometer had not yet been installed. In order to see the effect of solar radiation on the thermostat experiment, the outer brick temperature of the south-facing wall of the reference house was used to differentiate between sunny and cloudy days. On a sunny day, this temperature can rise 68°F (20°C) or more above the surrounding outdoor temperature. On a cloudy day, the brick temperature tracks the outdoor temperature within a few degrees. A difference of 68°F (20°C) or more between outdoor temperature and brick temperature was arbitrarily chosen to indicate sunny days, and a difference of less than 41°F (5°C) was arbitrarily chosen to indicate cloudy days. Days where the temperature difference fell between the high and low thresholds were labeled as mixed.

Drywall Temperature Measurement

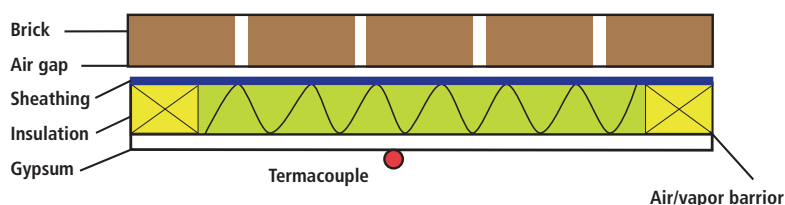


Figure 1. Cross-section of a wall, showing the location of the thermocouple (not drawn to scale).

Window Temperature Measurements

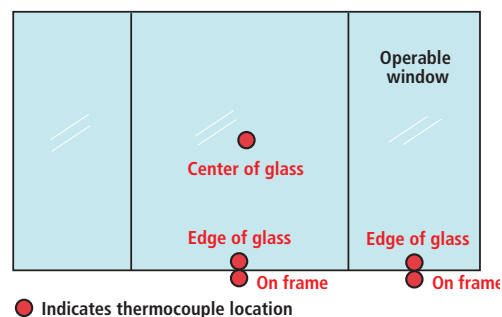


Figure 2. The thermocouples are located at different points on the inner surface of the window.

It should be noted that because of the nature of the data, summer data were divided into two groups (sunny and cloudy), while winter data were divided into three groups (sunny, cloudy, and mixed). Most winter days were either perfectly sunny or perfectly cloudy; there were only two days that could be classified as mixed. As a result, sunny and cloudy data formed two very distinct trends, with two days of mixed data clearly not belonging to either trend. In summer, there were very few completely sunny days or completely cloudy days. In essence, almost all of the summer days were mixed. For this reason, we chose a threshold to separate mixed days into two trends: cloudy and sunny. As mentioned above, we used a pyronometer to measure vertical solar radiation in the summer, with a threshold of 8.5 megajoules per square meter (MJ/m²) to distinguish between sunny and cloudy days. As expected, given the mixed summer weather, resultant summer trends show more scatter than winter trends.

Benchmarking

To compare the performance of the houses, we plotted daily energy consumption. During the heating season, the mid-efficiency furnaces in the two houses consumed almost the same amount of gas. Unfortunately, they did not consume the same amount of electricity. This resulted from differences in the power draw (wattage) of the two motors at heating speed in the twin houses. These power differences are due partly to static-pressure differences in the ductwork, which place different loads on the two motors, and partly to differences inherent in the motors themselves.

Similar trends are apparent in the summer benchmarking results. During the 2003 cooling season, on-times and air conditioner consumption were similar for the two houses. However, differences in furnace fan motor performance resulted in higher electrical consumption for the test house.

Care should be taken in applying the results of this study to other homes. Thermostat setback savings will be different for different houses and

different mechanical setups. Furthermore, the CCHT houses have different features than many houses, and these differences mean that the results of the study might not always apply to other homes. The CCHT houses are built to R-2000 standards; therefore, they hold heat better than older houses. As a

result, they don't cool down as quickly during setback for example, and there is less benefit to the strategy. This was seen in warmer weather, where savings were negligible. Also, during thermostat setback, lower quality windows and insulation could lead to lower surface temperatures and additional condensation problems.

Do Thermostat Adjustments Save Energy?

The winter experiments demonstrated that setting back the thermostat during the day and night saved energy in the CCHT test house (see Tables 1 and 2). As the setback temperature decreased, savings increased. Also, higher savings (expressed as a percentage) were



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The Twin Research House facility study showed that setting your thermostat ahead in the summer saves energy.

achieved on colder days, with longer furnace on-times.

A night and daytime setback of 64°F (18°C) reduced the length of time the furnace ran, resulting in furnace fan electrical savings of up to 6.4% and furnace gas consumption savings of up to 17% on the coldest day. A night and daytime setback of 61°F (16°C) saved up to 8.1% and 21% in electrical and gas consumption respectively. On warm or sunny days, the heating demand is less, and so savings were reduced. Projecting these results to the entire heating season revealed furnace gas seasonal savings of 13% with the 61°F (16°C) day and night setback, and 10% with the 64°F (18°C) day and night setback. Predicted furnace fan electrical savings were lower for the

Table 1. Predicted Winter Gas Savings from Thermostat Setback in the CCHT Test House

	Benchmark	64°F Night Setback	64°F Night and Day Setback	61°F Night and Day Setback
Furnace gas consumption (MJ/yr)	66,131	61,854	59,231	57,241
Savings from benchmark (%)	—	6.5	10.4	13.4

Table 2. Predicted Winter Electrical Savings from Thermostat Setback in the CCHT Test House

	Benchmark	64°F Night Setback	64°F Night and Day Setback	61°F Night and Day Setback
Winter furnace fan electrical consumption (kWh/yr)	2,314	2,295	2,270	2,261
Savings from benchmark (%)	—	0.8	1.9	2.3

Table 3. Summer Thermostat Experiments

Condition	Total Furnace and A/C Electric Consumption (kWh)	Predicted Savings from Benchmark (%)
72°F (22°C) Summer Benchmark	3,099	—
77°F (25°C) Day Setforward	2,767	11
77°F (25°C) Day Setforward – all sunny	2,690	13
77°F (25°C) Day Setforward – all cloudy	3,010	2.9
75°F (24°C) Higher Setting	2,376	23

season: 2.3% and 1.9% savings for the 61°F and 64°F night and daytime setbacks respectively.

Recovery times from thermostat setback were all less than two hours, and were usually less than one hour. The lower the temperature the house was allowed to reach (the lower the thermostat setback setting), the longer the recovery time. Because of this effect, the thermostat setback temperature and setback times should be chosen wisely to ensure occupant comfort in the early morning, after the nighttime setback, and during the early evening, after the daytime setback. Settings should be anticipated to allow the house ample time to reheat to a comfortable temperature before the occupants get up or return home from work.

Measured drywall surface temperatures remained above 55°F (12.7°C) for the 61°F setback, and above 64°F (17.8°C) for the 64°F setback. (It should be noted that surface temperatures were measured at the center of an insulated wall cavity. Lower surface temperatures could be expected on the wall stud framing, at the bottom plates, at corners, or in sections with poorer thermal characteristics.) Based on these measurements, no drywall condensation problems would be expected in the CCHT test house during the setback experiments, unless RH levels were above 55%. Based on current Health Canada recommendations, the Canada Mortgage and Housing Corporation suggests that RH be kept between 30% and 50%, and at 30% when exterior temperatures are below 14°F (-10°C). Excesses in humidity can lead to window condensation, stains on walls and ceilings, structural damage, and mold, which can cause allergic reactions. Humidity levels fluctuate with the number and activities of the occupants. Breath and perspiration, cooking, showering, bathing, and washing can all increase humidity levels in the home.

Window surface temperatures in both houses, on the other hand, were problematic. The frame of the window reached temperatures as low as 27°F (-2.6°C), even under benchmark conditions, with no setback. We would expect this to lead to condensation and ice problems on the frame, unless RH levels were kept below 19%—not a comfortable level—and below the

thermostat to a higher temperature proved to be a more effective summer strategy than employing a daytime setforward. During the cooling experiments, the higher temperature setting produced consistently greater savings than the setforward strategy, as would be expected—furnace circulation fan and A/C electrical savings of 23% for the cooling season were calcu-

lated, based on monitored results. Not only did the higher temperature setting produce similar savings on cool and hot summer days, but savings were not reduced on cloudy days. The drawback of the higher temperature setting is that, unlike the setfor-

ward strategy, it slightly increases indoor humidity. This increase in both temperature and humidity could make the house less comfortable for the occupants. **H_e**

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For more information:

A full report on the project described in this article is available from the Canadian Centre for Housing Technology (CCHT). Go to www.ccht-cctr.gc.ca.

Thermostat setforward produced savings of up to 21% in A/C electrical consumption and 5.2% in furnace fan consumption in the test house on the hottest, sunniest day.

CMHC recommended 30%.

Thermostat setforward savings increased with outdoor temperature and solar gains (see Table 3). Thermostat setforward produced savings of up to 21% in A/C electrical consumption and 5.2% in furnace fan consumption in the test house on the hottest, sunniest day, totaling over 6.3 kWh electrical savings for that day. Unfortunately, the energy savings from setforward were offset by poor recovery time—up to seven hours for these same hot days—the same length of time as the setforward itself. This could make the house very uncomfortable for the occupants during hot summer evenings.

Thermostat setforward savings were substantially reduced on cloudy days. If all days during the summer were cloudy, we would expect test house electrical savings of only 2.8% on A/C and furnace electrical consumption, as opposed to 13% electrical savings for a completely sunny summer.